

Opening a business in one of the major cities in the United States or Canada

Introduction/Business Problem

In the IBM Data Science certification program in the Data Science Capstone course we made clusterization of the New-York and Toronto areas (Old Toronto) and learned how to work with the Foursquare API.

The target audience of our project are potential investors in the United States or Canada.

In this project we chose opening a business (presumably a restaurant business) in one of the largest cities in the United States or Canada. It is necessary to justify the choice of the most suitable city from the point of view of the demographic situation. It is planned to study the city neighborhoods in more detail and determine the most suitable neighborhoods of chosen city for opening a particular type of business (type of restaurant). We want to find if the needs for ethnic restaurants are sufficiently represented for each of most presented groups or is there a shortage of them.

Data

1. Information about the demography of cities World Population Review [1].
2. Data on the structure of the city of Toronto from Wikipedia:
 - 2.1. Administrative structure of Toronto [2]).
 - 2.2. Address Data of Toronto districts [3]
3. The location Data of the districts was obtained using the Geocoder Python package.
4. Data of Toronto venues was obtained using the Foursquare API service.
5. Data on the ethnic composition of Toronto from the site Statistics Canada [4]

Methodology

In Part 1 we analyzed the five largest cities in the United States and Canada in order to select the city to open our business. To do this, we extracted information about the demographics of the world's largest cities from [1]. Data in table form was extracted from the site page using the Python library BeautifulSoup. Major cities in the United States and Canada were selected. The total number and dynamics of growth were compared.

In Part 2 we compiled an array of data about the addresses and neighborhoods of all boroughs of the city we selected. To find district addresses, we used data from Wikipedia [3]. The Geocoder python library was used to find GEODATA. Data were visualized and marked in a different colors (as belonging to a specific borough)

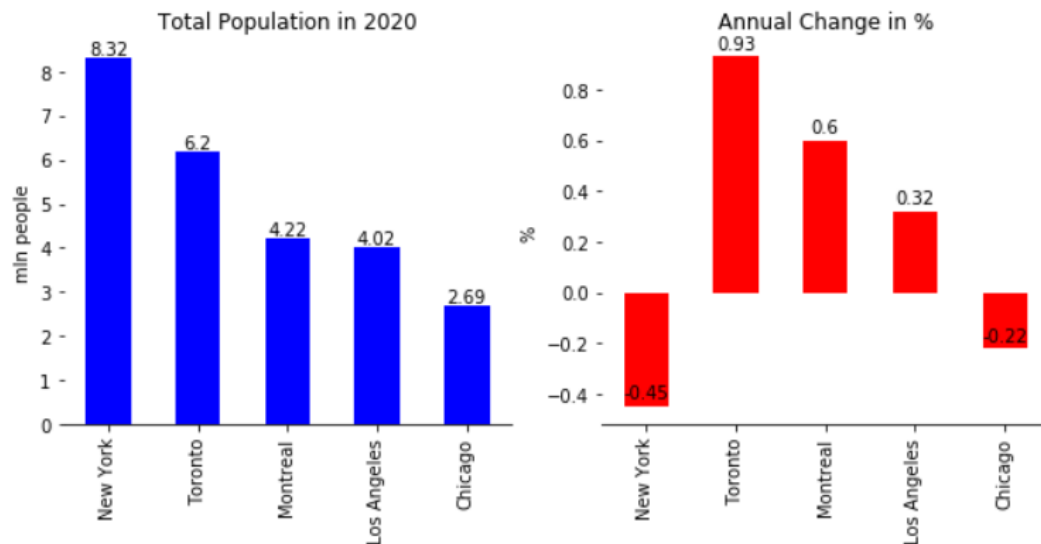
In Part 3 from the data [4], approximate information was obtained about the number of five major ethnic groups in the selected city. We use values of "Total - the language most often spoken at home" .

In Part 4 using the Foursquare API service, we received data about venues in the city of our choice. The data was sorted by venue category. Restaurant categories were selected. Further, based on the already obtained data on the ethnic composition, an assessment was made of how well a particular group is provided with a restaurant type that is more suitable for its ethnic characteristics. The assessment was carried out in comparison with other selected groups.

In Part 5 based on the data we received about venues, we selected the most visited categories of places next to which, in our opinion, it would be advantageous to place restaurants. Then machine learning clusterization methods were used to identify features. It is planned to allocate the most suitable areas for opening a business of a particular type (a certain type of restaurant)

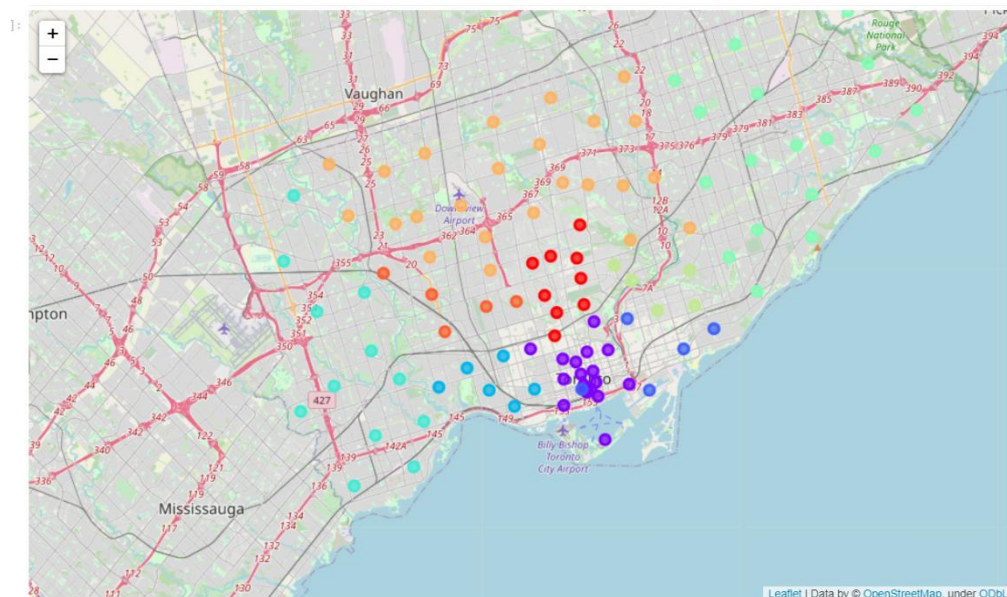
Results

The following data shows the population and annual changes in the population of five major cities in the United States and Canada.



As we can see, despite the largest population among the Top 5 Cities, new York has a negative population growth, in contrast to Toronto, which has a minimal figure among the top five cities! That's why Toronto was selected as the city for a more detailed study.

The following are neighborhoods of the city of Toronto grouped by borough.

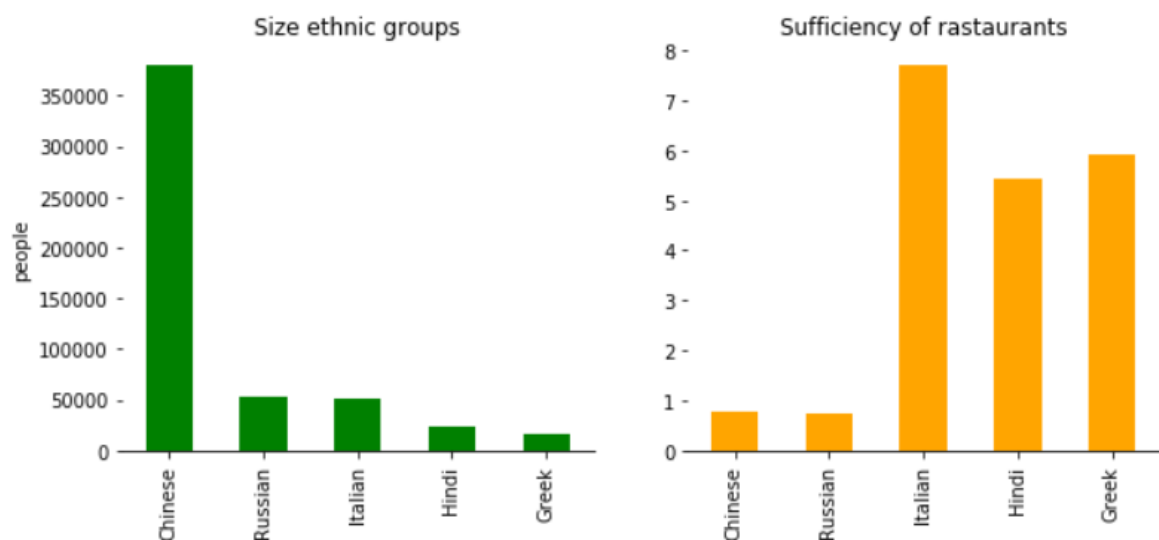


The table shows the number of selected ethnic groups based on values of "Total - the language most often spoken at home"

Language	Total	Restaurant Count	Proportion
Chinese	380550	30	0.788333
Russian	53675	4	0.745226
Italian	51815	40	7.719772
Hindi	23995	13	5.417795
Greek	16940	10	5.903188

According to the Foursquare API in Toronto we found 12 Chinese 18 and Asian restaurants. We took it in Chinese group. But there is no Russian restaurants at all! So for Russian group we took Eastern European restaurant (4).

Here we have it on barcharts:

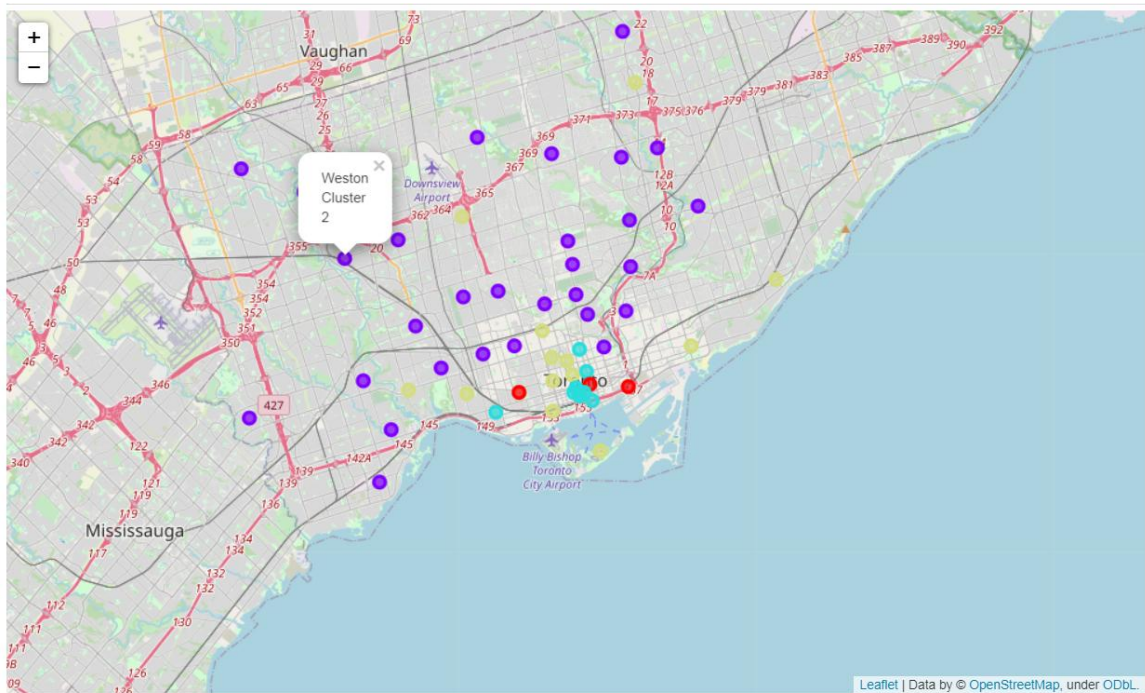


Discussion

At least two conclusions can be drawn from this data:

- There is still a shortage of chinese restaurant. But you can still visit Korean (5) and Thai (26) restaurants
- There is a clear lack of options for representatives of the Russian group. We didn't find a single Russian restaurant, and we took the closest ones - Eastern European restaurants in the statistics.

We have selected the most suitable venues categories in our opinion next to which our restaurant could be located such as colleges, parks, gardens, hotels, museums, theaters, concerts. We have divided districts into 4 clusters by the most common venue category.



As we can see, clusters 1 (red dots) and 3 (cyan dots) are mostly concentrated in the Central part of Toronto, so in our opinion they are the most successful options for placing a restaurant in terms of the largest concentration of people. The most frequent venue categories in cluster 1 are parks, theaters, gardens and college buildings. Therefore, we believe that we should focus on young people and students in this cluster. In addition, frequent visitors to these places are both permanent residents and tourists. The most frequent venue categories in cluster 3 are hotels, concert halls, museums. Therefore, we believe that we should focus on tourists and visitors from other cities in this cluster.

Conclusion

So in that project we:

We rated Toronto as the most promising city in the coming years to open a business in it among the top five cities in the United States and Canada.

We conducted an ethnic analysis of five major language groups in Toronto.

We found that among the two ethnic groups there is the greatest need to represent their needs in the restaurant business.

We have identified the most successful areas for choosing a restaurant opening location.

This is mostly a demonstration of the learned material of the IBM Data Science Capstone course. A more detailed analysis is needed to make a specific financial decision!

Reference list:

- [1]. <https://worldpopulationreview.com/world-cities/>
- [2]. https://en.wikipedia.org/wiki/List_of_neighbourhoods_in_Toronto
- [3]. https://en.wikipedia.org/wiki/List_of_postal_codes_of_Canada:_M
- [4]. <https://www12.statcan.gc.ca/census-recensement/2016/dp-pd/prof/details/page.cfm?Lang=E&Geo1=CMACA&Code1=535&Geo2=PR&Code2=35&Data=Count&SearchText=Caledon%20East&SearchType=Begins&SearchPR=01&B1=All>